

Georgia Institute of Technology

CS 4365/6365: Introduction to Enterprise Computing

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Project Checkpoint 2 Report

Group: 16

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Project Name: ContractLens: A Privacy-Preserving Contract Intelligence Platform

Context and Related Work:

Contract review is a critical part of the business process used in vendor agreements, software licensing agreements, NDAs, employment contracts, and other commercial transactions. Legal teams spend an average of 3.2 hours manually reviewing every single contract, with complex agreements requiring 8-12 hours across multiple different reviewers [1]. This work is essential with research from world commerce and contracting indicating that companies lose roughly 9.2% of annual revenue due to poor contract management practices [2]. Before agreements are executed, legal and business teams must identify important clauses and assess risks which can be a lengthy process.

Commercial platforms such as Ironclad and other Contract Lifecycle Management (CLM) systems demonstrate the demand for AI contractual review to speed up the process. The Contract Understanding Atticus Dataset (CUAD) provides over 13,000 expert annotations across 510 commercial contracts and 41 clause categories which can enable the development and evaluation of contract-analysis systems. Recently the ACORD benchmark has highlighted the importance of contract clause retrieval and search of past cases in legal review.

While recent LLM tools can summarize documents and answer questions, much of the underlying contract-review workflow could benefit from more specialized document understanding capabilities such as validated risk scoring, clause classification, contract comparison, and semantic retrieval of relevant cases. ContractLens aims to address this by building a contract intelligence platform that transforms unstructured contracts into structured contract representations that support clause classification, risk analysis, contract comparison, and question answering. The system will make full use of the legal datasets available while supporting privacy preserving local deployment for companies that cannot share sensitive contracts with external AI services.

1. <https://www.cyberbase.ai/blog/contract-review-roi>
2. <https://stealthagents.com/research/ai-contract-review-automation-statistics-2026>

Project Plan (Plan)

Provide clear understanding of the current state of the respective idea/related work to provide context and the starting point.

The goal of ContractLens is to develop a privacy-preserving contract intelligence platform that assists with first-pass contract review. Rather than functioning just as a general purpose chatbot, the system will transform the unstructured contracts into structured representations that support clause classification, semantic retrieval, contract comparison, risk analysis, and question answering.

The system will allow its users to upload PDFs, DOCX, or text format which will be first parsed and segmented into individual clauses and sections. A clause classification component will be used to identify important contract categories such as confidentiality, termination, liability, and intellectual property. These structures representations will then be used downstream for retrieval, comparison, and risk analysis.

ContractLens will support natural-language interaction through an LLM interface, but the primary functionality will be implemented in specialized backend services mentioned earlier. MCP-style tool integrations will be used to expose these services to the LLM interface.

The project will leverage publicly available legal datasets including the Contract Understanding Atticus Dataset (CUAD) for clause classification and evaluation. Performance will be evaluated using standard metrics such as precision, recall, and F1 score for clause classification, as well as retrieval metrics such as Recall@K and Mean Reciprocal Rank (MRR). More evaluation will be done to focus on the quality of contract comparison results, risk identification, and overall contract review reports.

The final deliverable will be a fully functional contract intelligence platform capable of extracting contract structure, identifying key clauses, comparing contractual language, highlighting potential risks, and generating review reports while supporting local deployment for privacy sensitive environments.

Project Deliverables

Deliverable	Description	Technical Stack
Contract Upload and Structured Review Workspace	Users will upload a contract and they can see a breakdown of the document which includes contract type, key sections, identified clauses, parties, and important metadata. This will	FastAPI, PDF Processing Libraries, Document Parsing Tools

	serve as the foundation to feed into downstream tasks.	
Contract Clause Exploration	Users can browse automatically identified clauses such as confidentiality, liability, termination, indemnification, and intellectual property. The system will highlight relevant section and link them back to the original contract.	LegalBERT, HuggingFace Transformers, CUAD Dataset
Contract comparison assistant	Users can compare contracts with a company template or an older version. The system will identify changes in obligations and terms and present them to the user.	Embedding Models, FAISS/ChromaDB, Retrieval Pipeline
Chatbot interface	Users will be able to discuss with a local LLM using natural language about their contract, receive cited answers, retrieve similar contractual language from the model, and generate executive review reports summarizing key findings and risks.	Local LLM, MCP Tools, RAG

Project Milestones

Show the project plan and the planned work for each teammate and remove this line.

Check point	Milestone	Technical Scope & Deliverables	Work Splitup	Status
Check point 1	Project planning and system design	Finalize the project scope, key deliverables, project plan, and project moat. Create github repository	1 person team	Done

Check point 2	Contract ingestion and clause intelligence	Implement the contract upload and parsing pipeline, structured extraction, and clause visualization. Users should be able to upload and view identified categories.	1 person team	Done
Check point 3	Retrieval, Comparison, and risk analysis	Implement semantic retrieval for comparison, implement pipeline for contract comparison, and clearly design and implement risk reporting backed by supporting evidence.	1 person team	pending
Check point 4	Platform and evaluation	Design and build ContractLens UI. Integrate chatbot for intelligent querying. Evaluate classification and retrieval performance with benchmark datasets.	1 person team	pending

Current Progress Report (Match)

- Provide a report on the work done during the last 2 weeks.
- Compare the work done with the initially planned milestone chart.
- Mention what your planned work is for the next 2 weeks.
- Mention any changes to the original plan due to the findings over the past 2 weeks.

Work done in the last two weeks:

All the deliverables listed in the project milestones table were implemented and are running. The FastAPI service accepts any PDF/DOCX/TXT uploads, parses them, breaks them down into clauses, and classifies each clause into categories. There is both a structured JSON view of the finalized classes and an HTML view. The UI design will be finalized for Checkpoint 4.

I integrated the ai-suggestions given in a reference branch and continued to develop locally. I completed any deliverables still pending in my local work and the code provided in the ai-suggestions branch. I did the following engineering with the CP1 feedback in mind:

1. I implemented persistent storage with an SQLite-backed store so that ingested contracts can survive a restart of the application rather than having to re-upload and reindex. I decided to do this because encoders can sometimes take a while so having contracts saved after one time indexing can save some time.
2. In addition to the rule-based classifier provided in the ai-suggestions, I implemented and fixed a design issue with LegalBERT encoder. The scaffold that was provided in the ai-suggestions was designed around LegalBERT being a zero-shot classifier but LegalBERT is an encoder with no actual classification head. I caught this issue and redesigned it so that it embeds each clause and each category's keyword description then classifies via cosine similarity to the closest description. I have an idea for fine tuning some local models for some downstream tasks I think an interesting avenue to explore here is to add a classification head to the encoder then retraining via CE loss on PACE but this is only necessary as a supplement or if I discover that the redesign is insufficient in performance for downstream tasks.
3. Built an eval-harness based on CUAD. Based on the feedback I got from CP1 I created an evaluation harness for this part of the application. The harness samples random clauses from the CUAD clause-classification dataset then reports precision/recall/F1 after running on the project backend.
4. The metadata extraction handles more than just two-party contracts as well as 5 common contract types seen in the data.
5. I grew the test suite by adding in PDF/DOCX tests, metadata extraction, the DB store, the classifiers which are all passing.

Comparison against the milestone chart:

The checkpoint 2 milestone is fully satisfied. It is also exceeded with the eval harness validation of the classification mechanism rather than just making a rule-based guess and a persistent storage layer rather to allow saving data over server restarts.

Work for the next two weeks:

Building off of the LegalBERT work, semantic retrieval and contract comparison will be implemented. Risk analysis supported by evidence in the contract will also be implemented.

Changes to the original plan:

1. The LegalBERT classification has to be redesigned to work for an encoder rather than a classification model but it still provides the expected output
2. The CUAD dataset does not include the categories of "Confidentiality," "Indemnification," or "Force Majeure" but the other 7 in the project taxonomy were accounted and evaluated for in the classifier

Supporting Evidence (Factual)

- Add the URL to your GitHub project to demonstrate the progress claimed above. Please provide links for individual files or directories pointing to the work you have done over the past 2 weeks. This will help TAs and peer reviewers efficiently assess factual accuracy.
- Datasets would show metadata, followed by a sample of data items. Software platforms would show documentation, followed by a sample demo (Hello World). Your own code would show a code listing, followed by execution results.
- Add screenshots as necessary.

Repo: <https://github.com/NB670/ContractLens>

Links to the specific files evidencing the work described above:

- app/store.py - persistent SQLite-backed contract store
- app/clauses/classifier.py - corrected embedding + cosine-similarity LegalBERT backend
- app/pipeline.py - expanded contract-type and multi-party detection
- scripts/generate_cuad_sample.py / data/cuad_sample.json - CUAD sample fixture (350 labeled clauses)
- scripts/evaluate_clauses.py - evaluation harness
- tests/ - full regression suite (30 tests)

Tests passing:

```
[(base) neelb@ipsec-10-2-0-188 ContractLens % python -m pytest -q
.....
30 passed in 1.02s
```

Eval harness results:

=== RuleBasedClassifier ===				
category	precision	recall	f1	support
Assignment	0.909	0.800	0.851	50
Governing Law	0.524	0.220	0.310	50
Intellectual Property	0.804	0.820	0.812	50
Liability	0.906	0.580	0.707	50
Payment Terms	0.875	0.140	0.241	50
Termination	0.818	0.900	0.857	50
Warranty	0.939	0.620	0.747	50
macro_avg	0.825	0.583	0.646	350

=== LegalBertClassifier ===				
category	precision	recall	f1	support
Assignment	0.000	0.000	0.000	50
Governing Law	0.980	1.000	0.990	50
Intellectual Property	0.854	0.820	0.837	50
Liability	0.838	0.620	0.713	50
Payment Terms	0.512	0.840	0.636	50
Termination	0.417	1.000	0.588	50
Warranty	0.750	0.060	0.111	50
macro_avg	0.622	0.620	0.554	350

LegalBERT is stronger in Governing Law but weaker elsewhere, this suggests that maybe the training of LegalBERT with the classification head could be worth it.

ContractLens 0.2.0 OAS 3.1

[/openapi.json](#)

Privacy-preserving contract intelligence platform (CP2: ingestion + clause intelligence)

default

GET	/	Root	⌵
POST	/upload	Upload Contract	⌵
GET	/contracts/{contract_id}	Get Contract	⌵
GET	/contracts/{contract_id}/view	View Contract	⌵

Schemas

Body_upload_contract_upload_post >	Expand all	object
HTTPValidationError >	Expand all	object
ValidationError >	Expand all	object

endpoints for upload, retrieval, and clause visualization.

sample_contract.txt

Type: Non-Disclosure Agreement · **Parties:** Acme Corporation, Globex LLC · **Clauses:** 7

Identified categories

- **Confidentiality:** 2
- **Governing Law:** 1
- **Indemnification:** 1
- **Intellectual Property:** 1
- **Liability:** 1
- **Termination:** 1

Clauses

MUTUAL NON-DISCLOSURE AGREEMENT [Confidentiality · 1.00]

chars 0-187

MUTUAL NON-DISCLOSURE AGREEMENT

This Mutual Non-Disclosure Agreement ("Agreement") is made and entered into by and between Acme Corporation and Globex LLC, dated as of January 1, 2026.

Clause 2 [Confidentiality · 1.00]

chars 187-444

1. Confidentiality

Each party agrees to hold in strict confidence all Confidential Information and proprietary information disclosed by the other party, and shall not disclose such trade secret information to any third party without prior written consent.

Clause 3 [Termination · 1.00]

chars 444-652

2. Term and Termination

This Agreement shall remain in effect for a period of two (2) years. Either party may terminate this Agreement upon thirty (30) days written notice of termination to the other party.

Clause 4 [Intellectual Property · 0.91]

chars 652-891

3. Intellectual Property

All intellectual property, including any copyright, patent, or trademark rights in the Confidential Information, shall remain the sole property of the disclosing party. No license is granted under this Agreement.

Sample contract after pipeline

Skill Learning Report

Mention each skill, followed by 1-2 lines on how you are making progress towards learning the skills or implementing them in your project, or a 3-4 line explanation on how you have learnt a particular skill.

Legal Document Understanding

I implemented and tested a clause segmenter and classifier that splits real contract text into pieces and classifies them. Learning how to do this for legal documents contributed to this skill.

Information Extraction

I built and tested a pipeline that takes in raw PDF text and produces a structured Contract model with typed clauses, offsets, contract type, and party lists.

Semantic retrieval

This is not yet a focus for checkpoint 2 but will be heavily developed in checkpoint 3

NLP in the legal domain

I built a real eval harness to validate the results of the clause classifier.

Local AI systems

Verified a real local HuggingFace transformer model downloaded to my computer for clause classification.

Self-Evaluation

1. Plan: 115

- a. Goes beyond yet another by focusing on converting unstructured text into something that can be molded to any task. Also takes privacy into consideration with local models which can be tuned to specialize in the task at hand. It doesn't use any APIs, it runs on-device which protects privacy.

2. Match: 119

- a. Everything claimed to have been done for checkpoint 2 was completed but also an eval harness to get real validation of the pipeline and models was created and so was persistent storage with a DB store.

3. Factual: 100

- a. All claims are backed in the factual section.

LLM Feedback (Claude Sonnet):

1. Project Plan (Plan): 113 / 120

Plan is unchanged from Checkpoint 1 and remains well-scoped; this checkpoint's execution (catching the LegalBERT architecture mismatch) validates the stack was genuinely understood, not just named.

2. Progress Report (Match): 118 / 120

Fully satisfies the Checkpoint 2 milestone and exceeds it with persistent storage, a corrected/verified ML classifier, and a real CUAD-based evaluation added in direct response to CP1 feedback.

3. Supporting Evidence (Factual): 97 / 100

Repository is live, correctly linked, and verifiable end-to-end: real commit history, real passing test suite, a real per-category precision/recall/F1 table for two classifier backends, and real screenshots. This is a substantive step up from Checkpoint 1's evidence.

Actionable Suggestions:

1. Reuse the persistent SQLite store as the backing layer for Checkpoint 3's retrieval index rather than building a separate one.
2. Decide whether to fine-tune LegalBERT or keep the rule-based classifier as default, now that real eval data shows where each backend actually fails (Assignment, Warranty).
3. Extend evaluation methodology to the 3 CUAD-uncovered categories qualitatively, so classification coverage stays complete going into later checkpoints.
4. Carry the same "measure, don't assume" discipline into Checkpoint 3's retrieval/comparison metrics.
5. Keep treating fine-tuning as a later-checkpoint stretch goal per CP1's own guidance, prioritizing retrieval/comparison/risk analysis first.